Casting Defect Inspection

Technical assessment - measured results

All values below are measured from the saved model and split manifest.

Confusion matrix rows are true Normal/Defective; columns are predicted Normal/Defective: [[257, 5], [5, 448]]. True negatives: 257; false positives: 5; false negatives: 5; true positives: 448.

Validation selected C=0.1 and threshold=0.36. The objective maximizes defect F2 subject to at least 95% precision on validation. The constraint was met: True. The threshold was locked before test evaluation. The final model is not refit on validation.

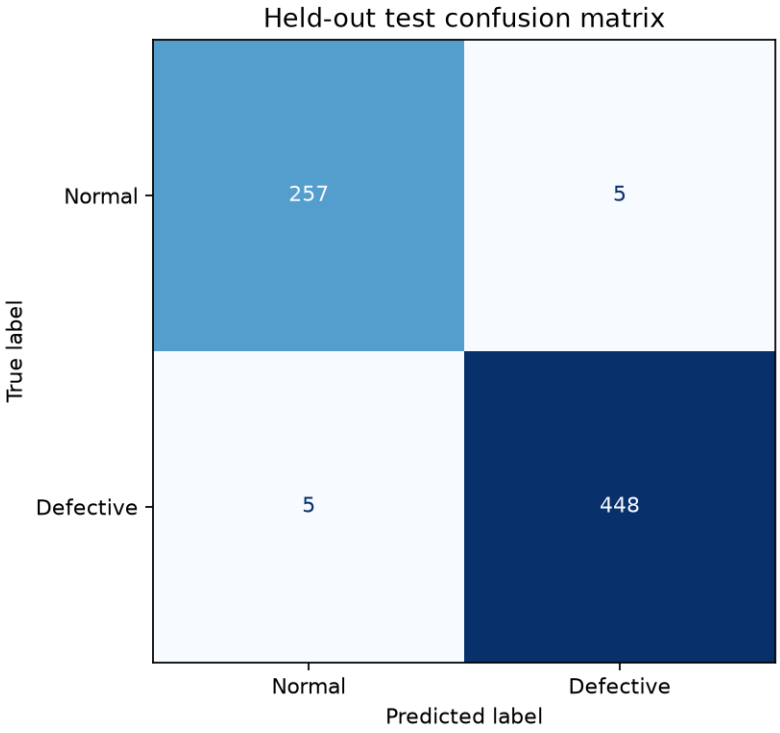
Training wall time: 272.3 seconds. Bootstrap 95% intervals (conditional on fixed predictions and image-level sampling): {"accuracy": [0.9762237762237762, 0.9944055944055944], "recall_defective": [0.9781181619256017, 0.997773198060587]}.

Split counts (0=Normal, 1=Defective): {"train": {"1": 3006, "0": 2249}, "validation": {"1": 752, "0": 562}, "test": {"0": 262, "1": 453}}. Unreadable images: 0; excluded exact duplicate train images: 64. Unique pixel hashes in the test set: 715. Potential test/train near matches (dHash <=4): 566. These are screening candidates; dHash similarity is not a reliable physical-part identifier. The 512px collection is excluded to avoid mixing unverified overlapping sources. The original 300px dataset was already augmented by its author. Exact deduplication does not prove independent parts across splits. A new part/batch/time-separated prospective dataset is required before a factory rollout.

Four real sample images returned HTTP 200 and matched local inference. A corrupt image returned HTTP 422. Full responses and request IDs: http_evidence.json.

Warm CPU inference service, two Torch threads, 50 measured requests after five warmups: HTTP p50 28.8 ms, p95 79.0 ms. Decode/preprocess/inference p50 26.2 ms, p95 75.2 ms. Hardware/platform: Intel64 Family 6 Model 158 Stepping 10, GenuineIntel, Windows-10-10.0.19045-SP0. This is a sequential localhost benchmark, not a throughput or SLA claim.

Metric	Held-out test
Accuracy	98.60%
Defect precision	98.90%
Defect recall	98.90%
Defect F1	0.9890


Held-out test confusion matrix
Normal
Defective
True label
257
5
5
448
Normal
Defective
Predicted label